13 (a) Show that the equation $x^4 - 3x - 1 = 0$ has a solution between $x = 1$ and $x = 2$

Let $f(x) = x^4 - 3x - 1$.

$f(1) = 1 - 3 - 1 = -3 < 0$

$f(2) = 16 - 6 - 1 = 9 > 0$

f is continuous, so the sign change shows

a root between 1 and 2.

1 mark
Substitute both values.

1 mark
Opposite signs.

(2)

(b) Show that for $x > 0$ the equation $x^4 - 3x - 1 = 0$ can be arranged to give $x = \sqrt[4]{3x + 1}$

$x^4 - 3x - 1 = 0 \rightarrow x^4 = 3x + 1$

Take the positive fourth root ($x > 0$).

1 mark
Correct rearrangement.

$x = \sqrt[4]{(3x + 1)}$

(1)

(c) Starting with $x_0 = 1$, use the iteration formula $x_{n+1} = \sqrt[4]{3x_n + 1}$ once to find an estimate for a solution of $x^4 - 3x - 1 = 0$

First iterate: $\sqrt[4]{(3 \times 1 + 1)} = \sqrt{2} = 1.414213562...$

Estimate after one iteration ≈ 1.414

1 mark
Use x = 1 once.

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(1)

(Total for Question 13 is 4 marks)